

Welcome to DATA 202!

K Arnold

Opening Prayer

From the apostle Paul's letter to the Philippians:

This is my prayer:
that your love may abound more and more
in knowledge and depth of insight,
so that you may be able to discern what is best and may be pure
and blameless for the day of Christ, filled with the fruit of
righteousness that comes through Jesus Christ—to the glory and
praise of God.

What is Data Science?

What is Data Science?

- **Data**: information, collected systematically
- **Science**: systematic study of that data

What is this course?

- **visualization**: communicating data to humans
- **modeling** and validation: using data to perform tasks
- but first, **data wrangling**: How to transform data into the structure you need for visualization and modeling

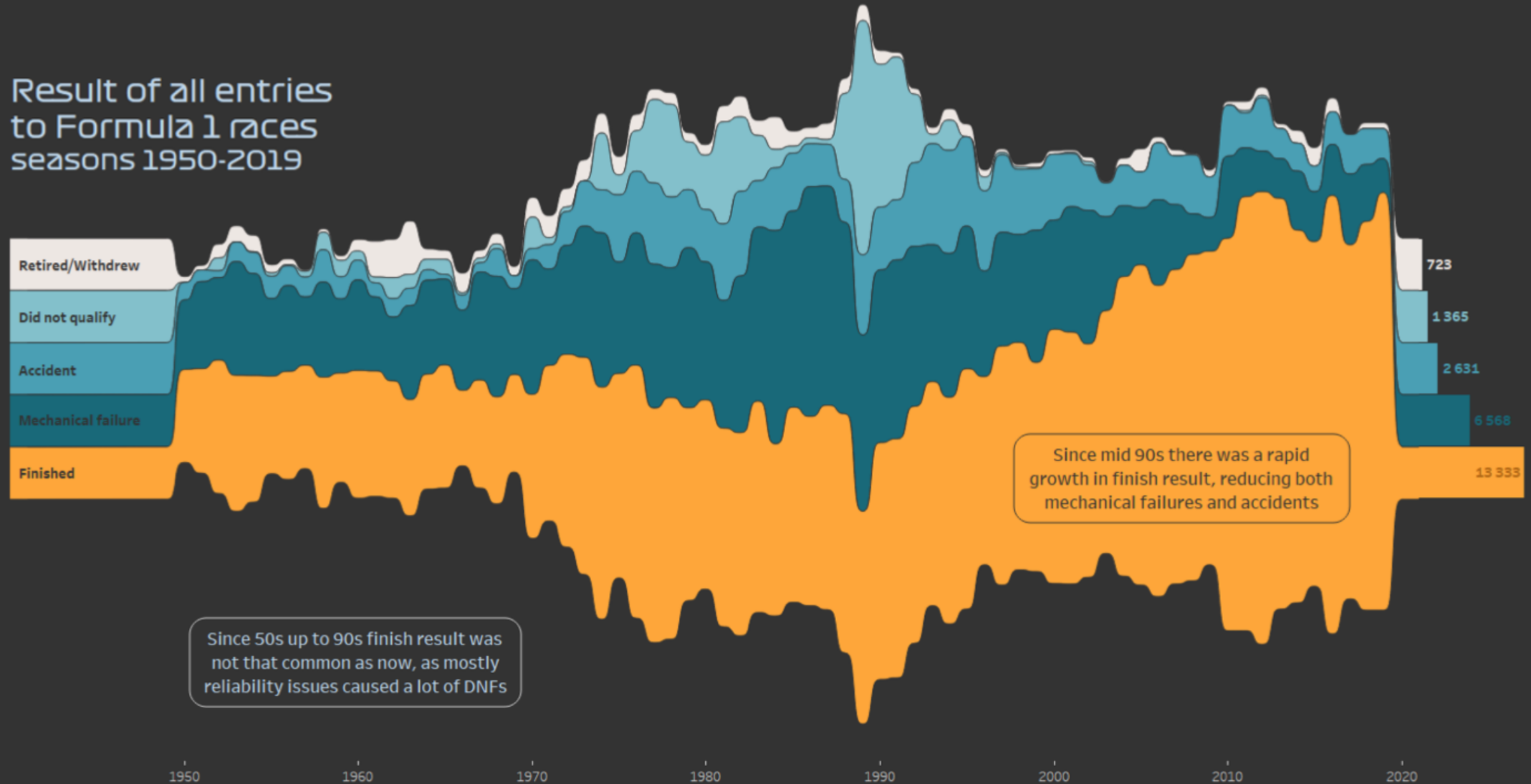
...using **computing** in the R language (`#rstats`) and occasionally other tools

Where are we going?

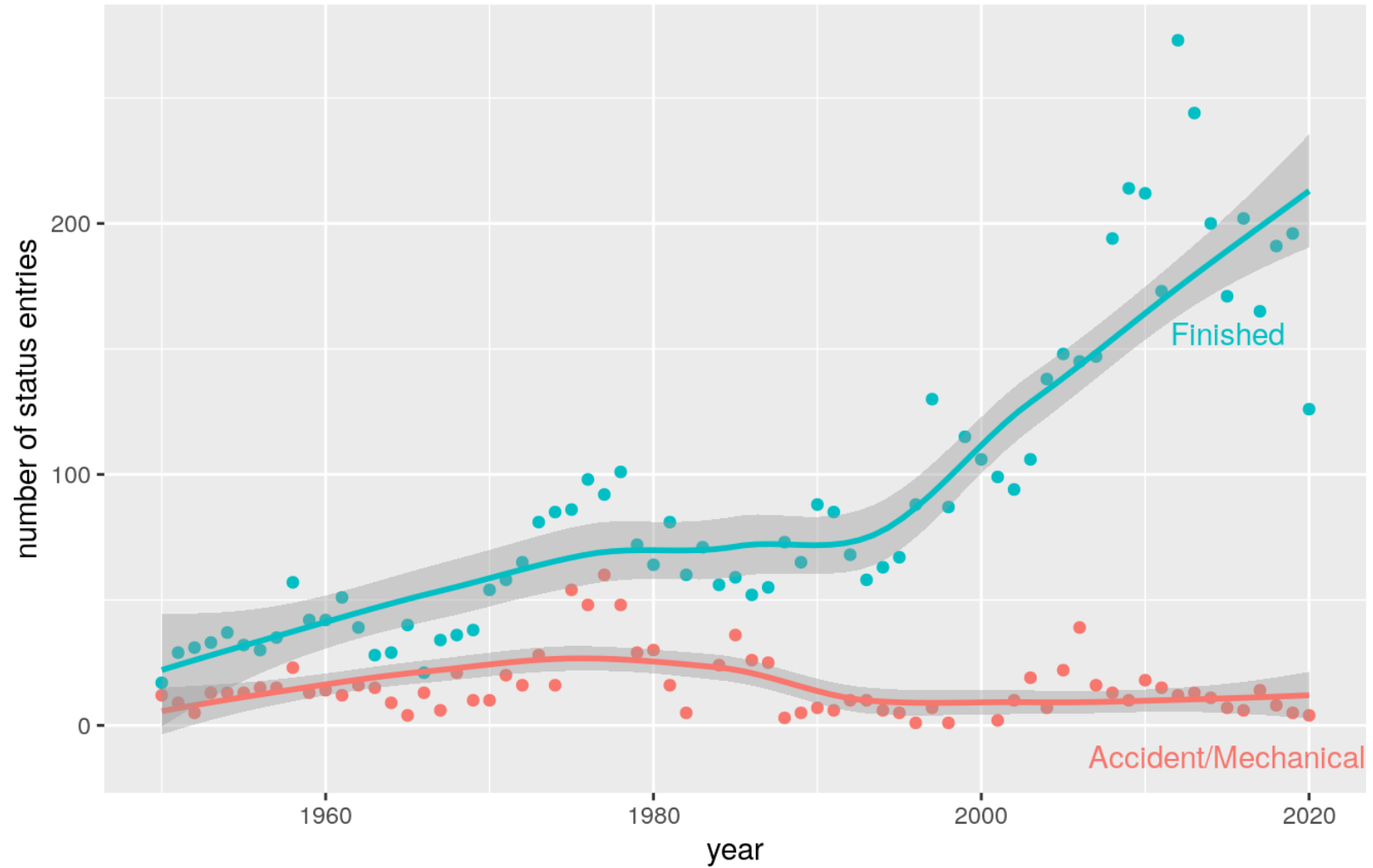
- Visualization Overview
- Visualization Details
- Wrangling (single-table)
- Wrangling (multiple tables)
- Tidy data
- Applications and Ethics
- Statistical Foundations Review
- Supervised Learning using Trees
- Unsupervised Learning
- Databases and APIs
- Other Supervised Learning Methods
- Parameter Tuning and Overfitting
- Text as Data
- Geospatial Data

F1 journey to current safety standards was not easy one

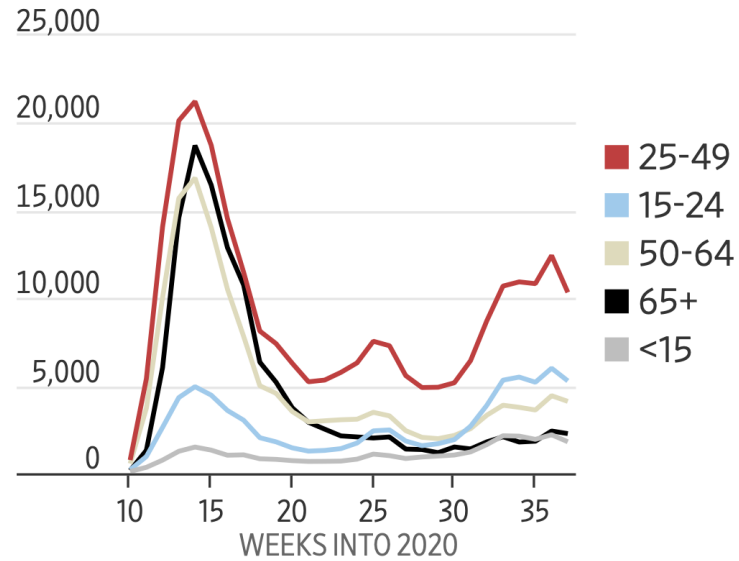
Result of all entries
to Formula 1 races
seasons 1950-2019



Finished vs. Failure Results of Formula 1 Race Entries from 1950-2019

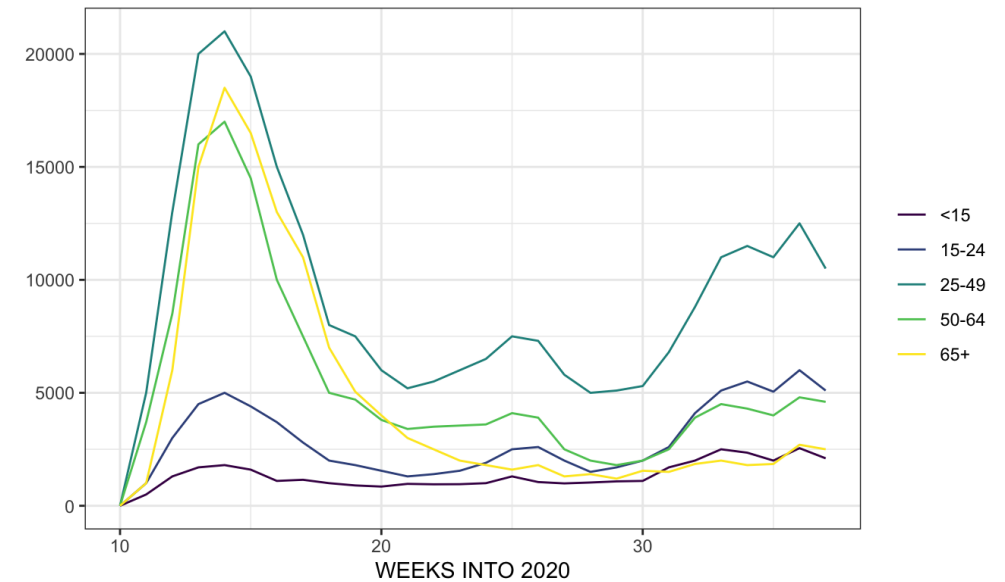


Confirmed Covid-19 cases in 17 European countries by age group



Note: Countries include Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, Germany, Iceland, Ireland, Latvia, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal and Sweden
 Source: European Surveillance System, part of the European Center for Disease Prevention and Control

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Example final projects

- Predict how much a used car will sell for
- Forecast how much electricity will be used
- Predict how much a plane flight will cost

Our Goals

- *Skill*: how to do these things
- *Knowledge*: understanding the underlying concepts
- *Dispositions*: wisdom in practicing these skills

Data Dispositions

Humility

Challenge: data feels powerful, people listen to what you use it to say.

So we will practice:

- Citing all sources (for both data and process)
- Acknowledging limitations
- Noticing and reporting our analysis decisions and possible alternatives
- Validation of results

Integrity

It's tempting to say something that isn't entirely true, or to manipulate the collection/analysis/reporting process to yield the answer you want.

So we will practice:

- Evaluating claims that others use data to make
- Clearly articulating our analysis decisions and rationale
- Reproducibility
- Using exploratory analytics to validate data against assumptions

Hospitality

We can choose to use our tools to elucidate and clarify, rather than obscure.

So we will practice:

- Clear visual communication
- Clarity of code and process
- Writing explanations that are accessible and appropriate to audience.

Compassion and Justice

Data Science can both cause harm and reveal it.

So we will:

- Study examples of how data might cause harm
- Study examples of how harm might be mitigated or revealed

A worked example

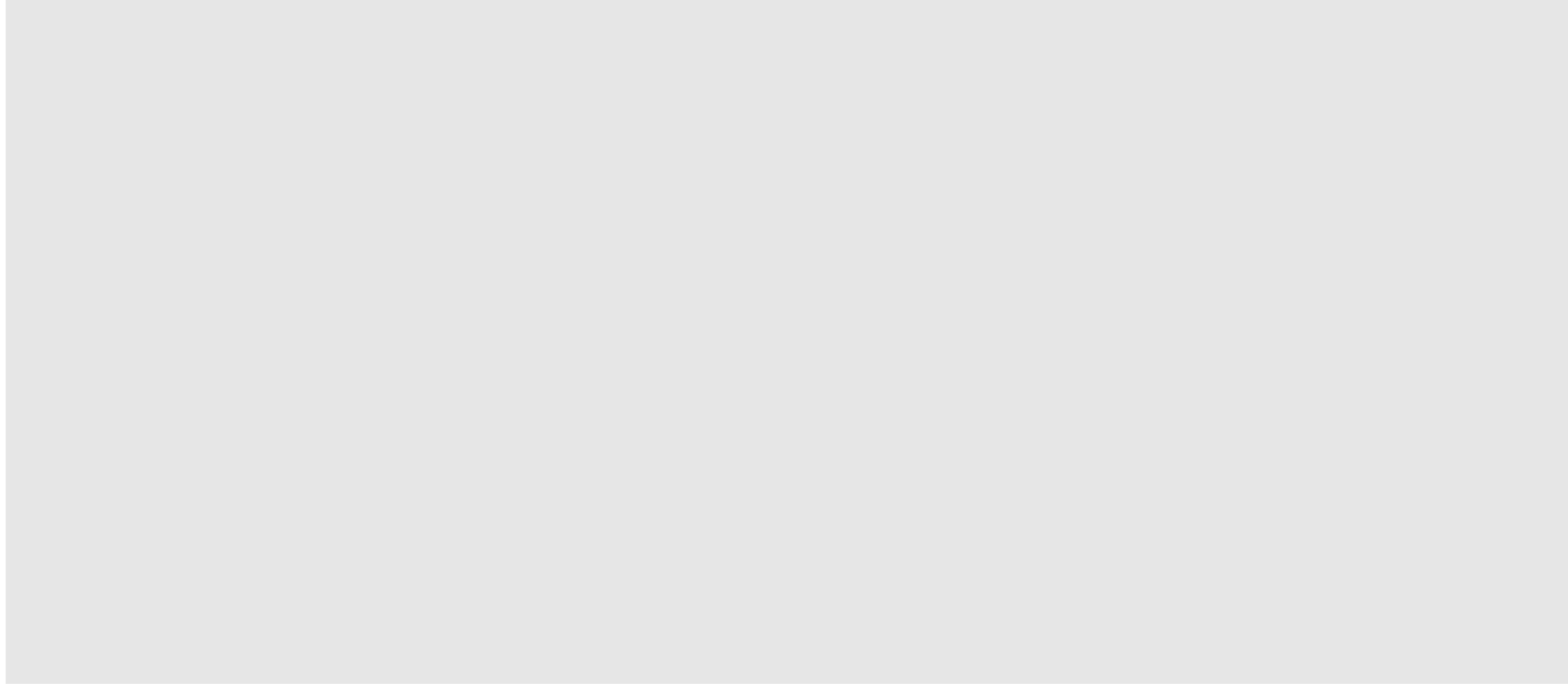
```
un_uk_us_tr
```

```
## # A tibble: 1,212 × 4
## # Groups:   country, year [219]
##   country year issue          percent_yes
##   <chr>   <dbl> <fct>          <dbl>
## 1 Turkey  1946 Colonialism      0.8
## 2 Turkey  1946 Economic development 0.6
## 3 Turkey  1946 Human rights      0
## 4 Turkey  1947 Colonialism    0.222
## 5 Turkey  1947 Economic development 0.5
## 6 Turkey  1947 Palestinian conflict 0.143
## # ... with 1,206 more rows
```

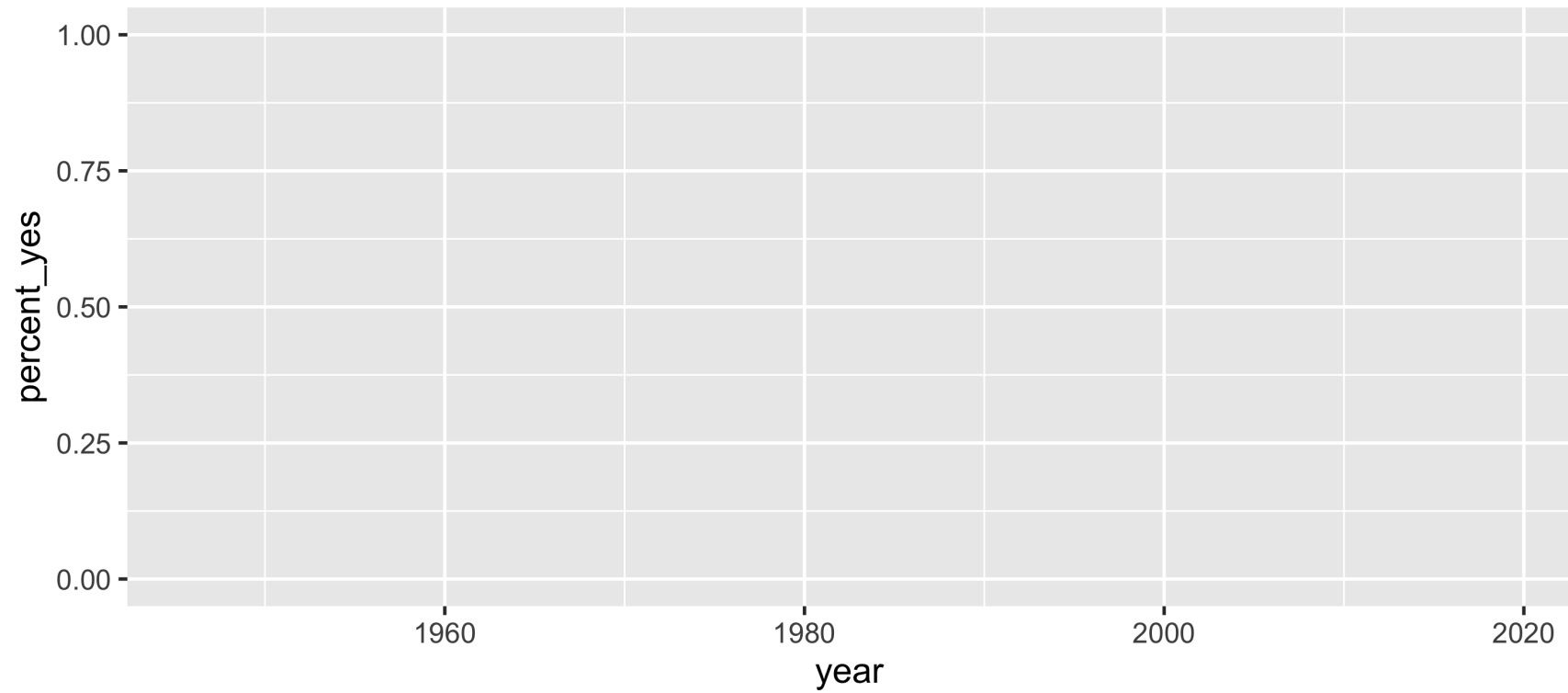
Erik Voeten "Data and Analyses of Voting in the UN General Assembly" Routledge Handbook of International Organization,

edited by Bob Reinalda (published May 27, 2013)

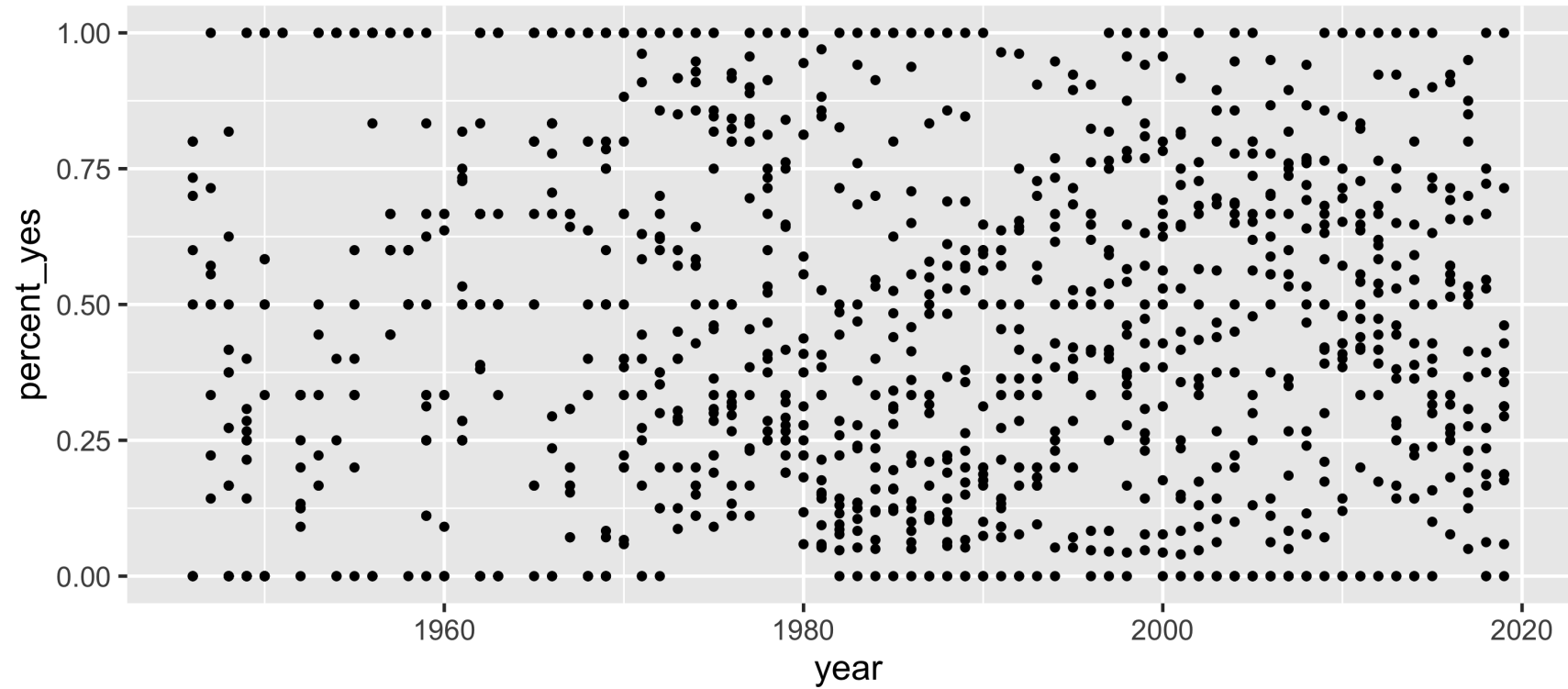
```
ggplot(un_uk_us_tr)
```



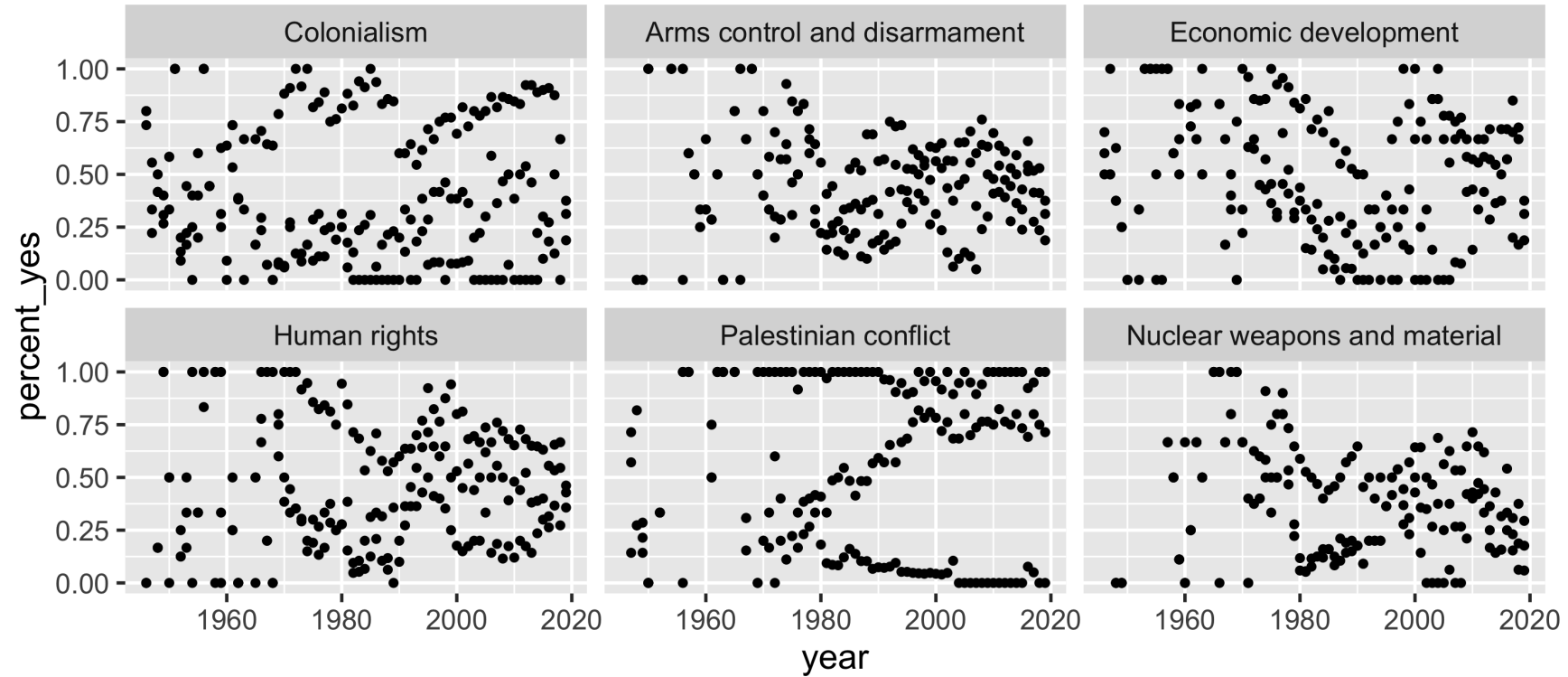
```
ggplot(un_uk_us_tr,  
       aes(x = year, y = percent_yes))
```



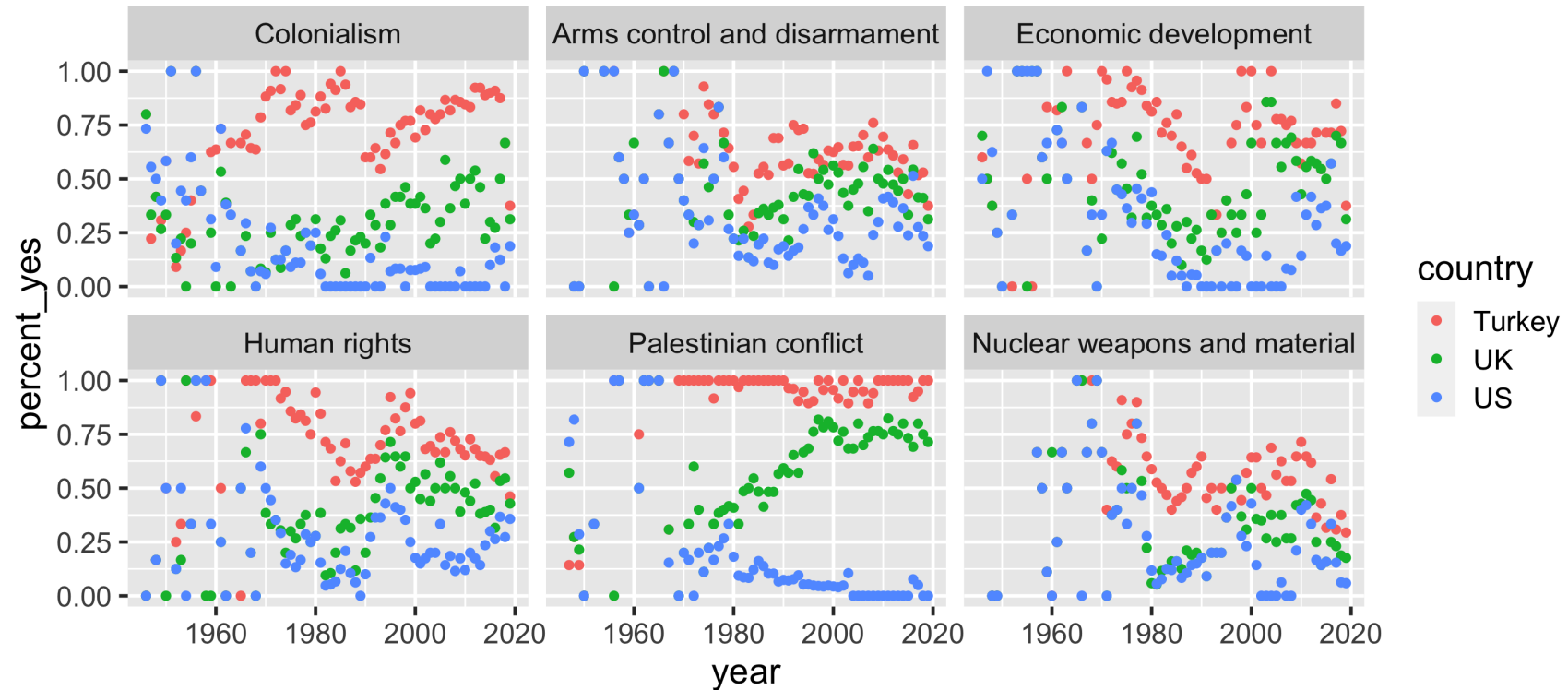
```
ggplot(un_uk_us_tr,  
       aes(x = year, y = percent_yes)) +  
  geom_point()
```



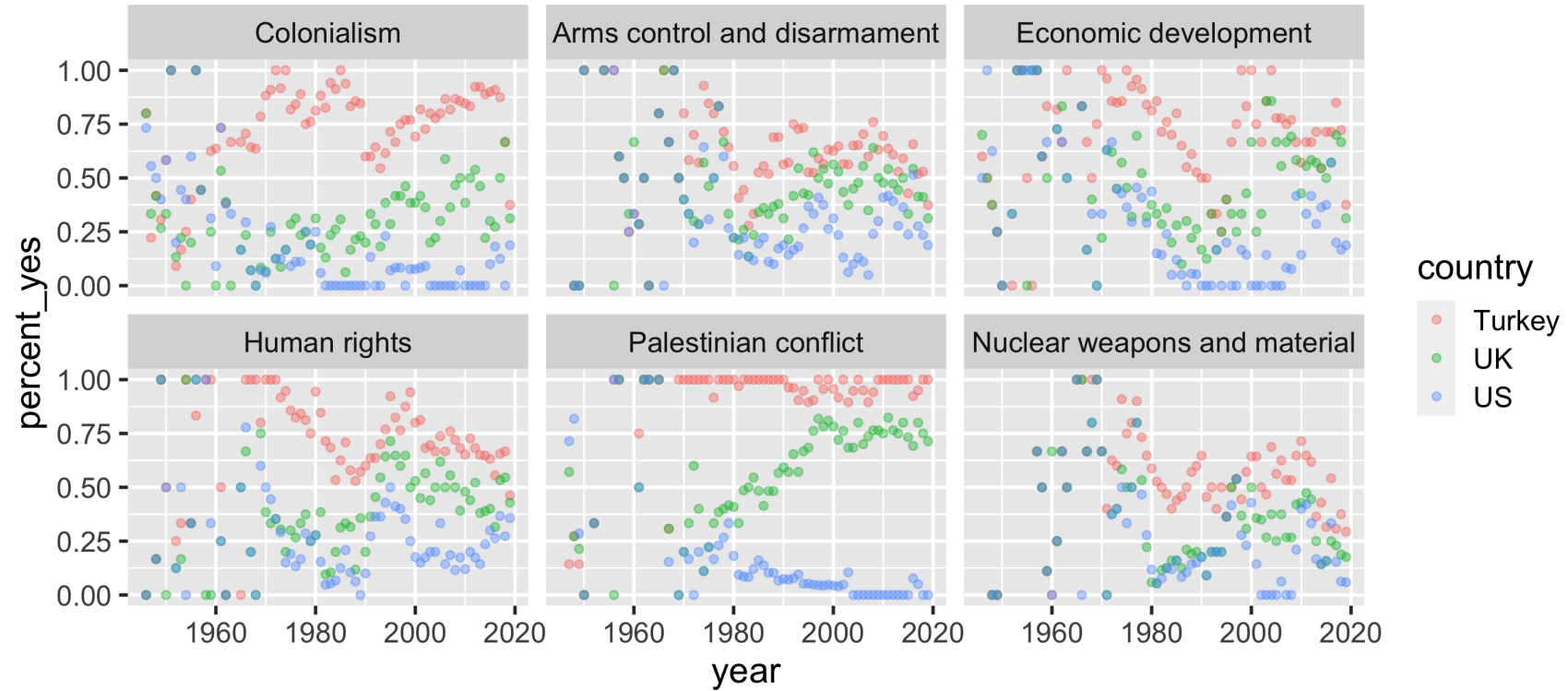
```
ggplot(un_uk_us_tr,  
      aes(x = year, y = percent_yes)) +  
  geom_point() +  
  facet_wrap(~issue)
```



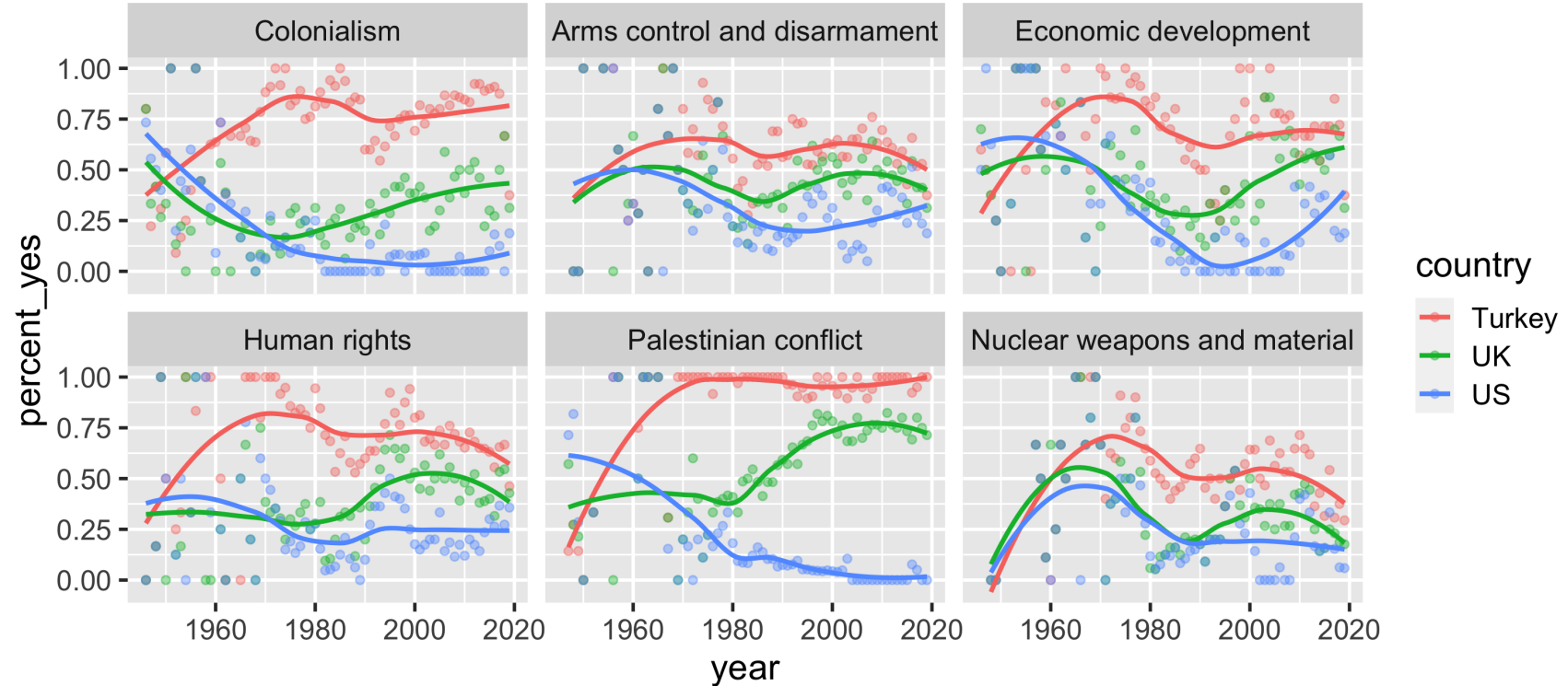
```
ggplot(un_uk_us_tr,
      aes(x = year, y = percent_yes, color = country)) +
  geom_point() +
  facet_wrap(~issue)
```



```
ggplot(un_uk_us_tr,
      aes(x = year, y = percent_yes, color = country)) +
  geom_point(alpha = 0.4) +
  facet_wrap(~issue)
```

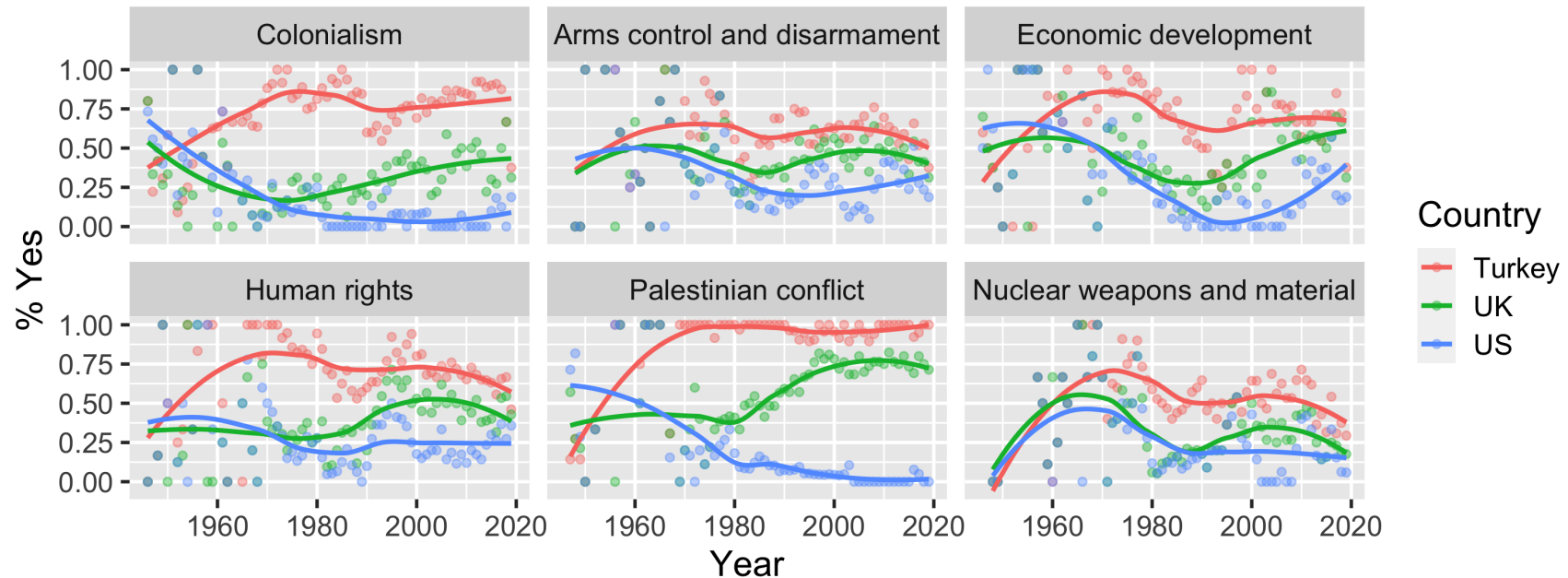


```
ggplot(un_uk_us_tr,
      aes(x = year, y = percent_yes, color = country)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "loess", se = FALSE) +
  facet_wrap(~issue)
```



```
ggplot(un_uk_us_tr,
      aes(x = year, y = percent_yes, color = country)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "loess", se = FALSE) +
  facet_wrap(~issue) +
  labs(
    title = "Percentage of 'Yes' votes in the UN General Assembly",
    subtitle = "1946 to 2015",
    y = "% Yes", x = "Year", color = "Country"
  )
```

Percentage of 'Yes' votes in the UN General Assembly
1946 to 2015



```
ggplot(un_uk_us_tr,
      aes(x = year, y = percent_yes, color = country)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "loess", se = FALSE) +
  facet_wrap(~issue) +
  labs(
    title = "Percentage of 'Yes' votes in the UN General Assembly",
    subtitle = "1946 to 2015",
    y = "% Yes", x = "Year", color = "Country"
  ) +
  scale_y_continuous(labels = label_percent())
```

Percentage of 'Yes' votes in the UN General Assembly
1946 to 2015

